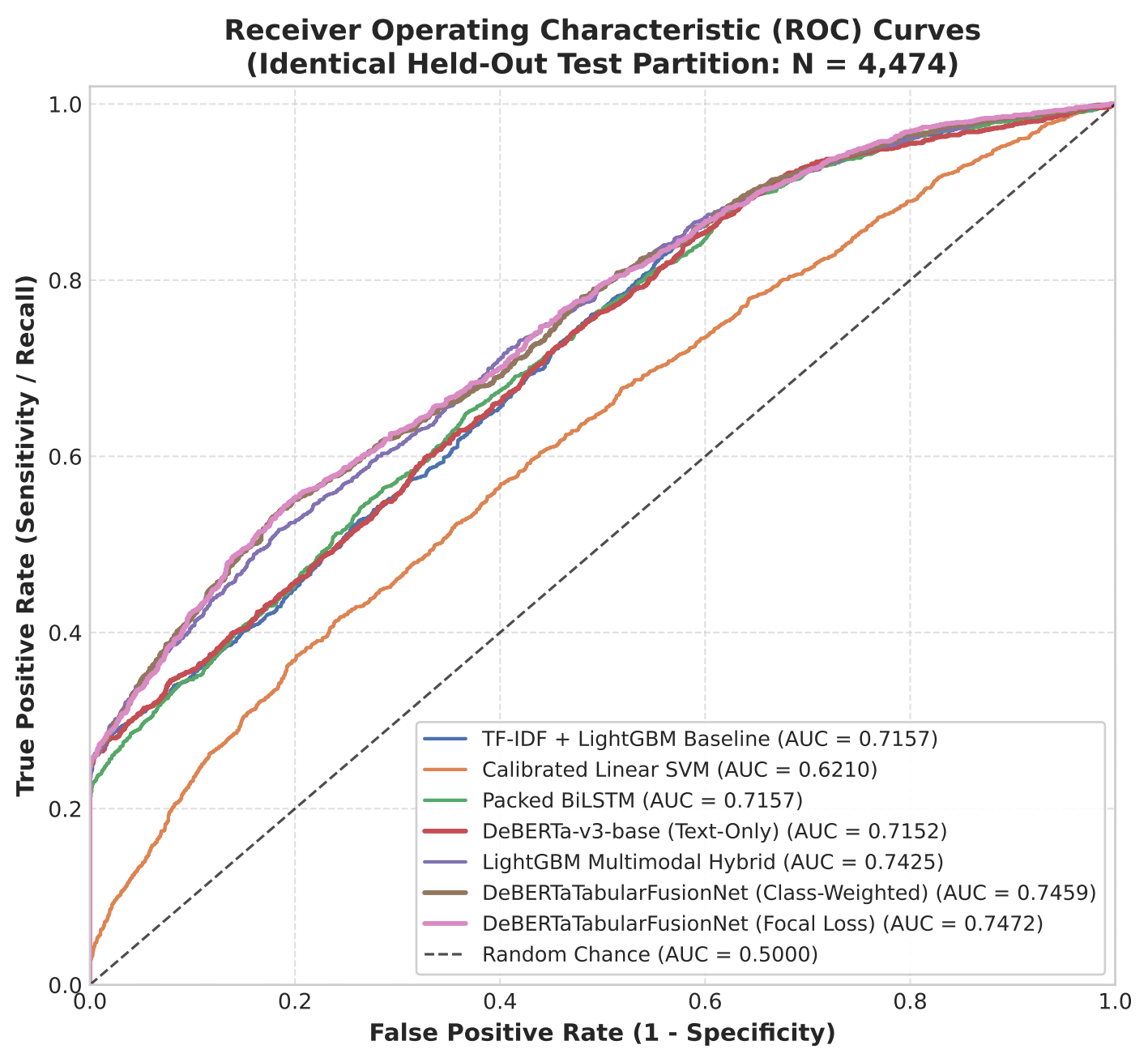

Receiver Operating Characteristic (ROC) Curves
(Identical Held-Out Test Partition: N = 4,474)
True Positive Rate (Sensitivity / Recall)
False Positive Rate (1 - Specificity)
0.0
0.2
0.4
0.6
0.8
1.0
TF-IDF + LightGBM Baseline (AUC = 0.7157)
Calibrated Linear SVM (AUC = 0.6210)
Packed BiLSTM (AUC = 0.7157)
DeBERTa-v3-base (Text-Only) (AUC = 0.7152)
LightGBM Multimodal Hybrid (AUC = 0.7425)
DeBERTaTabularFusionNet (Class-Weighted) (AUC = 0.7459)
DeBERTaTabularFusionNet (Focal Loss) (AUC = 0.7472)
Random Chance (AUC = 0.5000)